MIL-C-48082B(AR)
22 JULY 1981
SUPERSEDING
MIL-C-48082A(PA)
27 April 1973

MILITARY SPECIFICATION
CARTRIDGE, 40MM, HEDP, M433; SPITBACK ASSEMBLY FOR METAL PARTS AND LOADING, ASSEMBLING AND PACKAGING

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts and the loading, assembling and packing for one type of spitback designated as Spitback Assembly for use in Cartridge, 40MM, HEDP, M433.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-650	-	Explosive, Sampling, Inspection Testing
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1188	-	Commercial Packing of Supplies and Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8886351 - Spitback Assembly

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify "49 CFR 100-199 (latest revision)".)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly. The assembly shall comply with all requirements specified on Drawing (Dwg.) 8886351, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Loading composition. The loading composition at the loading station at the time of loading the spitbacks shall comply with the requirements specified on Drawing (Dwg.) 8886353 and with all requirements specified in applicable specifications.

3.4 Functioning. The spitbacks shall function satisfactorily as evidenced by the cup and liner loading assembly functioning high order.

3.5 First article testing. This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

3.6 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and quality assurance provisions Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are made a part of this detail specification.

4.2 Classification of inspection. The following type of inspections shall be conducted on this item:

a. First article inspection.
b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Liner, Spitback	8886352	25
Housing, Spitback	8886354	25
Spitback Assemblies	8886351	315

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078

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TABLE I. FIRST ARTICLE INSPECTION

MIL-C-48082B(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Spitback Assembly and Components				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Liner Spitback (Dwg. 8886352)				
	Examination of defects	25			4.4.2.1
	Housing Spitback (Dwg. 8886354)				
	Examination of defects	25			4.4.2.2
	Spitback Assembly (Dwg. 8886351)				
	Examination	25			4.4.2.3
	Functioning	315			4.5.4

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Each inspection lot shall contain:

a. Metal parts of one interfix lot number from one manufacturer.

b. Loading mixture consisting of batches blended by the same process. Each batch shall be composed of the same ingredients, each of which is to be from one interfix lot from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48082B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Liner Spitback	1 OF 1	8886352
			NEXT HIGHER ASSEMBLY: 8886351

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
		See 4.5.1			
CRITICAL:	None defined				
MAJOR					
101.	Outside diameter		0.40%	3.2	Gage
102.	Minimum wall thickness, after forming		0.40%	3.2	Gage
MINOR					
201.	Spherical radius		0.65%	3.2	Gage
202.	Height		0.65%	3.2	Gage
203.	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48082B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8886354
4.4.2.2	Housing Spitback				NEXT HIGHER ASSEMBLY 8886352
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Smallest outside diameter		0.40%	3.2	Gage
102.	Second largest inside diameter		0.40%	3.2	Gage
103.	Conc of second largest inside diameter with smallest outside diameter including perpendicularity with flange		0.40%	3.2	Gge
104.	Diameter counterbore		0.40%	3.2	Gage
105.	Concentricity of counterbore with second largest inside diameter		0.40%	3.2	Gage
106.	Distance from small end to flange		0.40%	3.2	Gage
107.	Dept of counterbore		0.40%	3.2	Gage
108.	Location small basic diameter		0.40%	3.2	Gage
109.	Location large basic diameter		0.40%	3.2	Gage
MINOR					
201.	Distance from large end to front face of flange		0.65%	3.2	Gage
202.	Thickness of flange		0.65%	3.2	Gage
203.	Diameter of flange		0.65%	3.2	Gage
204.	Large chamfer on flange missing		0.65%	3.2	Visual
205.	Body diameter		0.65%	3.2	Gage
206.	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48082B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8886353
4.4.2.3	Spitback Loading Assembly				NEXT HIGHER ASSEMBLY 8886351
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Depth to charge, small end		0.40%	3.2	Gage
102.	Charge above flush, small end		0.40%	3.2	Gage
103.	Charge above flush, liner end		0.40%	3.2	Gage
104.	Depth to charge, liner end		0.40%	3.2	Gage
MINOR					
201.	Caked explosive on liner assembly		0.65%	3.2	Visual
202.	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48082B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Spitback Assembly		8886351
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Length from forward flange to crimped end	125	0.40%	3.2	Gage
102.	Swell diameter		0.40%	3.2	Gage
103.	Lacquer missing		0.40%	3.2	Visual
104.	Functioning		0.40%	3.4	4.5.4
MINOR					
201.	Crimp not full 360 degrees		0.65%	3.2	Visual
202.	Evidence of poor workmanship		0.65%	3.6	Visual

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method Column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Dimensional control of stamped parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.5.2 Composition.

4.5.2.1 Determination of calcium resinate. Transfer an accurately weighed portion of approximately 5 grams (gm) of the sample to a tared medium porosity, sintered glass crucible. Extract the specimen with three separate 20 millilitar (ml) portions of chloroform which have been previously saturated with RDX, maintaining each portion in contact with the residue for approximately five minutes before appling suction. Stir occasionally to break up any lumps and wash down the sides of the crucible with each 20 ml portion of chloroform to insure complete removal of the calcium resinate.

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Aspirate the residue until te odor of chloroform is no longer detectable, place the crucible and contents in a drying oven at 100 degrees Centigrade (°C) to 105°C for one half hour, cool in a desiccator and weigh. Repeat the above extraction, drying and weighing procedures with invidudual 20 ml portion of chloroform until the loss in weight on subsequent extractions total less than 0.003 gm. Calculate the loss in weight of the crucible plus contents to percent calcium resinate on a mositure-free basis as follows:

$$\text{Percent calcium resinate} = \frac{100A}{W}$$

Where:

A = loss in weight of the crucible plus contents
W = weight of specimen on a moisture-free basis

4.5.2.2 Determination of the graphite content. Extract the residue, remaining after calcium resinate (See 4.5.2.1) has been removed with two 20 ml portions of dimethyl formamide, maintaining each portion in contact with the residue for approximately five minutes with intermittent stirring before applying suction. Wash the crucible and contents with a minimum of three 20 ml portions of acetone, maintaining each portion in contact with the residue for approximately one minute with continuous stirring before applying suction, until the following qualitative test for RDX in the acetone washings is negative. Transfer the acetone filtrate to a 25 ml beaker, evaporate to dryness on a steam bath and add several drops of a solution of chromotropic acid in 36 normal (N) sulfuric acid. The formation of a reddish brown color, as compared to a pink color, obtained by a blank prepared in an identical manner, indicates the presence of RDX. Aspirate the residue until the odor of acetone is no longer detectable, place the crucible and contents in a drying oven at 100 degrees to 105 degrees C for one half hour, cool in a desiccator and weigh. Calculate the weight of the residue in the crucible to percentage of lubricant in the sample on a moisture-free basis as follows:

$$\text{Percent lubricant} = \frac{100\ A}{W}$$

Where:

A = increase in weight of empty tared crucible
W = weight of sample on a moisture-free basis

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4.5.2.3 Determination of RDX. The percent RDX shall be determined by subtracting the percent calcium resinate and lubricant from 100 percent.

4.5.3 Moisture content of composition. The moisture content shall be determined using Karl Fischer method in accordance with Standard MIL-STD-650, Method 101.4 with methanol as the special solvent.

4.5.4 Functioning. The spitback assembly shall be assembled to an HE loaded cartridge for which it is standard. The cartridge shall be fired from a launcher for which it is standard to impact on a steel plate target located 125 feet from the launcher.

Observation shall be made for proper functioning (See 3.4). If fuze fails to function (dud), the test sample shall be disregarded. If more than four (4) fuze duds occur, replacement samples shall be fired.

5. PACKAGING

5.1 Unless otherwise specified, packaging shall be Level B.

5.1.1 Preservation and packing. Preservation and packing of the liner spitback and housing spitback for maximum and intermediate protection (Levels A and B) shall be in accordance with MIL-STD-1169. Minimum protection shall be in accordance with MIL-STD-1188.

5.1.2 Preservation and packing of the spitback assembly for shipment shall be in accordance with Title 49, Code of Federal Regualtions, Section 173.66.

5.2 Unitization.

5.2.1 Unitization of the spitback liner and spitback housing shall be in accordance with MIL-STD-1169.

5.2.2 Unitization of the spitback assembly shall be in accordance with instructions issued by the procuring agency.

5.3 Marking.

5.3.1 Marking of the spitback liner and spitback housing shall be in accordance with MIL-STD-1169.

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5.3.2 Marking of the spitback assembly shall be in accordance with MIL-STD-129 and Title 49, Code of Federal Regulations, Section 173.66.

5.4 Shipping. When components from more than one lot are shipped together, the lots shall be kept separated and the division between lots clearly inidcated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The assembly covered by this specification is intended for use with Cartridge, 40MM, HEDP, M433.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment design for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAA-I, Dover, NJ 07801. Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAA-S.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities of ARRADCOM.

6.6 Changes from previous issue. Asterisks are not used in this revision to identity changes with respect to the previous issue, due to the extensiveness of the changes.

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MIL-C-48082B Cartridge, 40MM, HEDP, M433; Spitback Assembly For Metal Parts

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